



MJ2386 Energy Storage Technology

Course responsible and Examiner: Justin Chiu

Battery Cell Discharge Performance Testing

Report by:

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1. Introduction

Electrochemical batteries are essential in modern life, enabling energy storage and powering a wide range of devices. Their operation is based on Reduction-Oxidation (Red-Ox) reactions within the battery cell. This experiment focuses on secondary batteries, which can be repeatedly charged and discharged. During discharge, electrons from the anode flow to the cathode through an external circuit, while positive ions move through the electrolyte. At the cathode, a reduction reaction occurs, completing the process and resulting in a measurable voltage drop. The charging process reverses these reactions, restoring the battery's energy and voltage.

Battery degradation is a key concern, leading to reduced capacity and performance over time. The main factors influencing degradation include cycling (repeated charge-discharge cycles), elevated temperatures (accelerating chemical reactions), operating beyond recommended limits (causing stress on electrodes and electrolytes) and improper storage conditions. These factors were central to our laboratory observations. [1]

In this experiment, we analyzed the discharge behavior of three battery types under varying conditions, including different C-rates and temperatures. The objective was to compare theoretical discharge characteristics with empirical results, providing insights into how operational factors influence battery performance.

Table 1: Data of the battery types used in the experiment

Battery Type	Nominal Capacity (mAh)	Termination Voltage (V)
Ni-MH	780	1
Li-Po	420	3
Lead-acid	2500	1.68

2. Methodology

This experiment investigated three key factors influencing battery discharge performance: C-rate, temperature, and battery chemistry. The tests were conducted under controlled conditions to ensure reliable results.

For **C-rate analysis**, Ni-MH batteries were discharged at room temperature with C-rates of 1 (Test 1.1), 0.5 (Test 1.2), and 2 (Test 1.3). The impact of **temperature** was assessed by discharging Ni-MH batteries at +40°C (Test 2.1) and -10°C (Test 2.2), both at a C-rate of 1. Lastly, **battery chemistry comparison** involved discharging a Lead-Acid battery at a C-rate of 2 (Test 3.2) and a Li-Po battery at a C-rate of 1 (Test 3.3), both at room temperature.

The procedure for all tests was consistent: batteries were fully charged and connected to a DC load terminal. For temperature-specific tests, terminals were preheated to +40°C or cooled to -10°C. Discharge parameters (current, voltage cutoff, and capacity) were configured, and data logging was initiated. The discharge process was monitored, and results were recorded after completion. This systematic approach ensured comprehensive analysis across all variables.

Table 2: Overview of the C-rate, temperature, and cell chemistry used in each test

Test	C-rate	Temperature	Cell Chemistry
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1.1	1	Ambient (Room Temp)	Ni-MH
1.2	0.5	Ambient (Room Temp)	Ni-MH
1.3	2	Ambient (Room Temp)	Ni-MH
2.1	1	+40°C	Ni-MH
2.2	1	-10°C	Ni-MH
3.2	2	Ambient (Room Temp)	Lead-acid
3.3	1	Ambient (Room Temp)	Li-Po

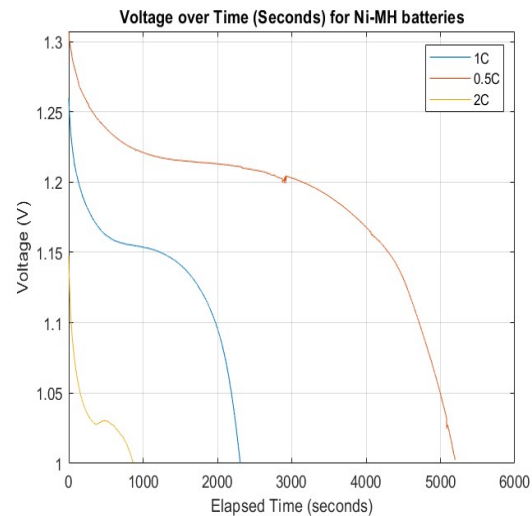
Throughout the process, we ensured the accuracy of the data and proper handling of the batteries.

3. Results and Analysis

In this section all graphs are presented that followed from the experiments with both the Ni-MH batteries as well as the Lead acid and Li-Po batteries.

3.1 Analysis of NiMH battery at ambient room temperature with different C rates

Voltage vs Time:



The graph illustrates the voltage over time for Ni-MH batteries discharged at three different C-rates: 0.5C, 1C, and 2C. Each curve demonstrates distinct discharge behavior based on the C-rate, highlighting the trade-offs between discharge speed and battery utilization:

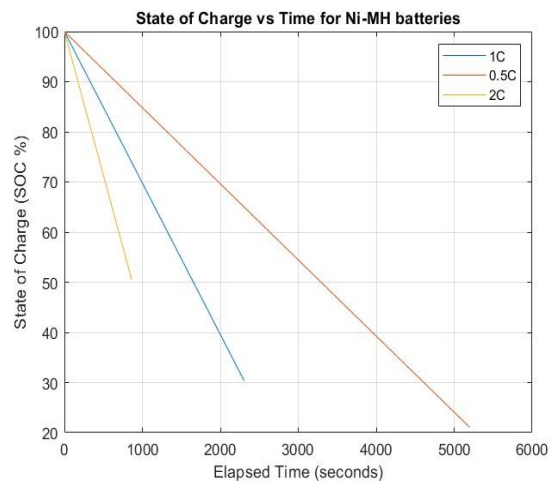
At **0.5C**, the discharge is the slowest, with the voltage maintaining a relatively steady plateau phase for an extended period before gradually decreasing. This indicates efficient energy usage and a prolonged discharge time.

At **1C**, the battery discharges faster compared to 0.5C. The voltage drop is steeper, but the plateau phase is still noticeable, showing a balance between discharge speed and energy utilization.

At **2C**, the discharge occurs rapidly, with a sharp voltage drop shortly after the test begins. The plateau phase is minimal, and the battery reaches its cutoff voltage quickly due to higher current demand and increased internal resistance.

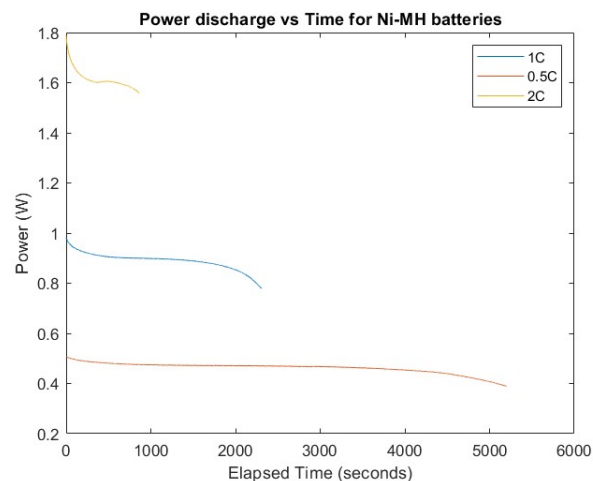
The trends emphasize that higher C-rates lead to quicker depletion of the battery's stored energy due to increased current demand, which also contributes to greater energy losses in the form of heat. Lower C-rates, on the other hand, allow for a more gradual and sustained energy delivery, maximizing the runtime of the battery.

State of Charge vs Time:



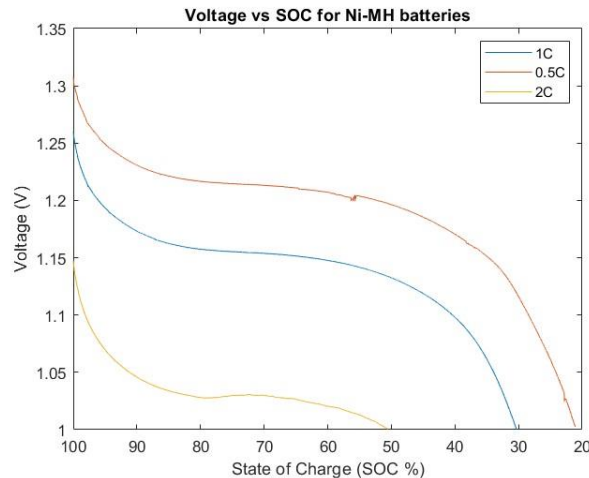
The graph illustrates **the State of Charge (SOC) over time for Ni-MH batteries discharged** at three C-rates: 0.5C, 1C, and 2C. At 0.5C, the SOC decreases gradually, indicating the longest discharge duration and efficient energy utilization. At 1C, the SOC declines at a moderate rate, balancing discharge speed and energy delivery. In contrast, at 2C, the SOC drops rapidly, reflecting the shortest discharge duration due to high current demand. This demonstrates that lower C-rates extend battery life by reducing the rate of energy depletion, while higher C-rates lead to faster energy exhaustion but quicker power delivery.

Power discharge vs Time:



The graph shows the **power discharge over time** for Ni-MH batteries at three different C-rates: 0.5C, 1C, and 2C. At 2C, the power starts at the highest value but decreases rapidly, reflecting a fast energy depletion rate. At 1C, the power output is relatively stable for a moderate duration before tapering off as the battery depletes. At 0.5C, the power discharge is the most consistent and steady over time, demonstrating efficient energy utilization and prolonged delivery. This highlights that higher C-rates deliver more power initially but at the cost of quicker depletion, whereas lower C-rates sustain power delivery for a longer time.

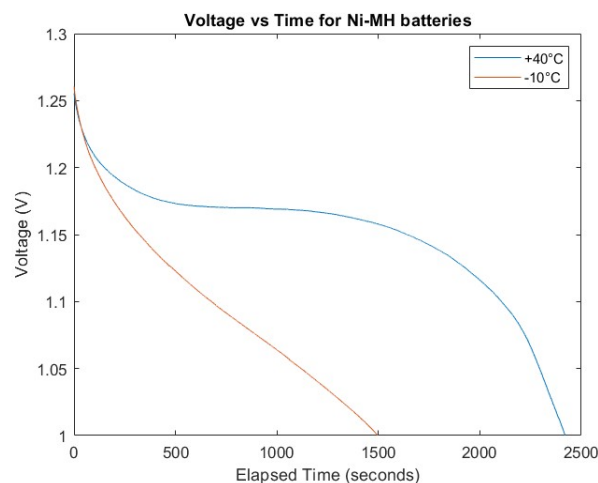
Voltage vs SOC:



The graph depicts the relationship between **voltage and State of Charge (SOC)** for Ni-MH batteries discharged at three different C-rates: 0.5C, 1C, and 2C. At 0.5C, the voltage declines gradually with SOC, maintaining a relatively high voltage for a longer duration, indicating efficient energy utilization. At 1C, the voltage decreases moderately, showing a balance between discharge rate and energy efficiency. At 2C, the voltage drops sharply even at higher SOC levels, reflecting rapid depletion and reduced energy efficiency. The trends demonstrate that lower C-rates sustain voltage levels for longer periods, while higher C-rates lead to quicker voltage and SOC depletion due to increased internal losses.

3.2 Analysis of NiCd battery at discharge at different temperature

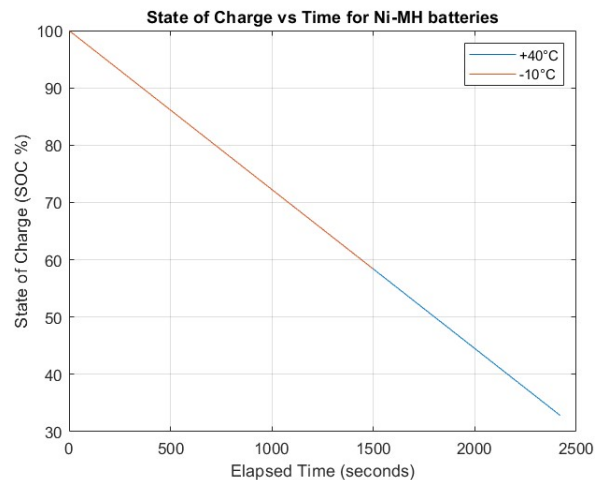
Voltage vs Time:



The graph compares the **voltage versus time** performance of Ni-MH batteries at two temperatures: +40°C and -10°C. At +40°C, the battery exhibits a slower voltage decline, maintaining a higher operational voltage

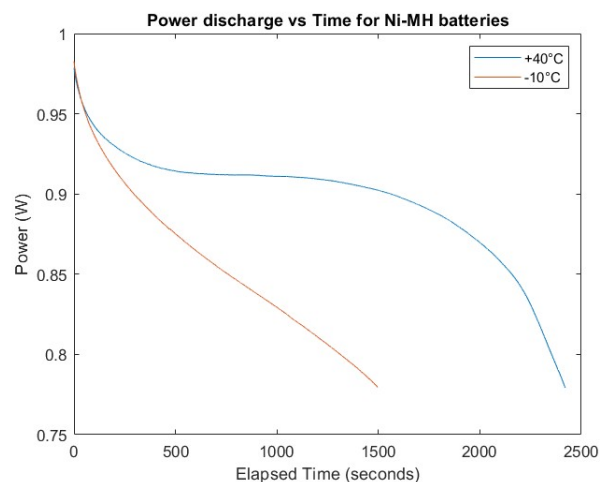
over time due to improved electrochemical activity and lower internal resistance at elevated temperatures. Conversely, at -10°C , the voltage decreases rapidly, reflecting reduced performance caused by slower reaction rates and higher internal resistance in colder conditions. This comparison highlights the significant influence of temperature on battery discharge efficiency and longevity.

State of Charge vs Time:



The graph compares the **State of Charge (SOC) over time** for Ni-MH batteries at two temperatures: $+40^{\circ}\text{C}$ and -10°C . At $+40^{\circ}\text{C}$, the SOC decreases more gradually, indicating a slower discharge rate and better performance due to enhanced electrochemical activity in higher temperatures. At -10°C , the SOC declines more rapidly, reflecting faster energy depletion caused by reduced reaction efficiency and increased internal resistance in colder conditions. This comparison highlights the impact of temperature on the battery's discharge behavior, with warmer temperatures supporting longer battery operation and colder temperatures limiting performance.

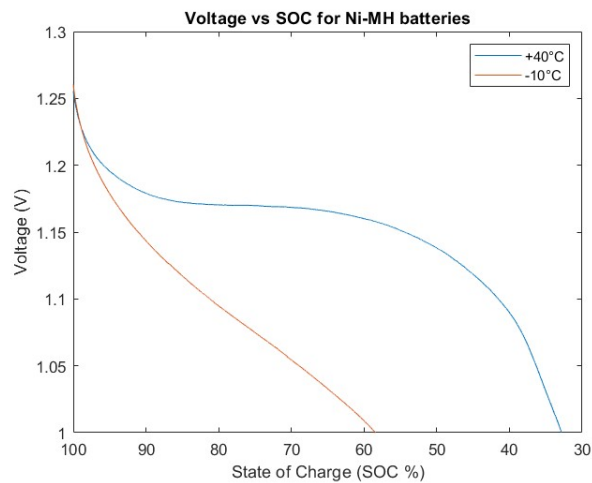
Power discharge vs Time:



The graph shows the **power discharge over time** for Ni-MH batteries at $+40^{\circ}\text{C}$ and -10°C . At $+40^{\circ}\text{C}$, the power remains higher for a longer duration and decreases gradually, reflecting improved battery efficiency due to enhanced electrochemical reactions at elevated temperatures. In contrast, at -10°C , the power output starts lower and declines more rapidly, indicating reduced efficiency and faster performance degradation.

caused by slower reaction rates and increased internal resistance in colder conditions. These results emphasize the significant impact of temperature on battery performance and energy delivery.

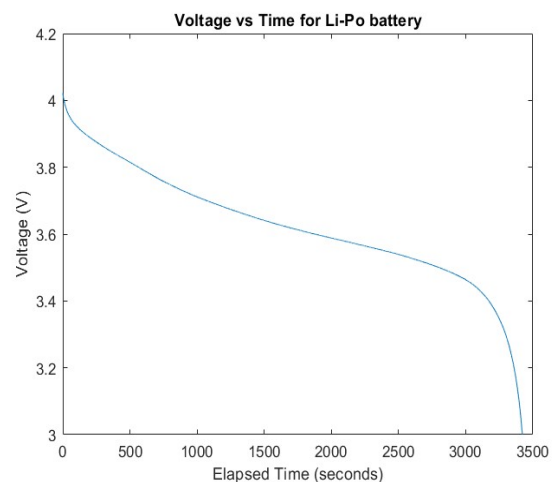
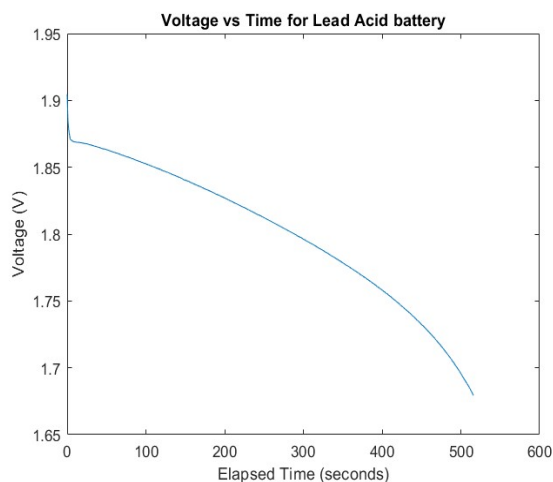
Voltage vs SOC



The graph illustrates the relationship between **voltage** and **State of Charge (SOC)** for Ni-MH batteries at two temperatures, +40°C and -10°C. At +40°C, the voltage remains higher throughout the SOC range and declines more gradually, indicating efficient energy utilization and stable performance due to improved electrochemical reactions at elevated temperatures. At -10°C, the voltage drops sharply even at higher SOC levels, reflecting limited capacity and reduced efficiency caused by slower reaction rates and increased internal resistance in colder conditions. The data highlights how temperature impacts the voltage-SOC profile, with higher temperatures supporting better energy output and lower temperatures leading to quicker voltage depletion.

3.3 Analysis of Lead Acid and LiPO battery cells

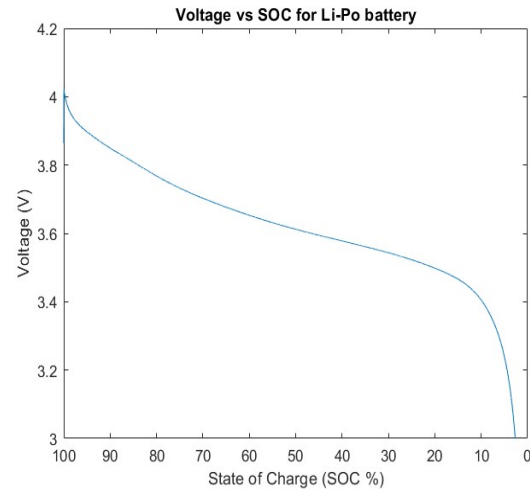
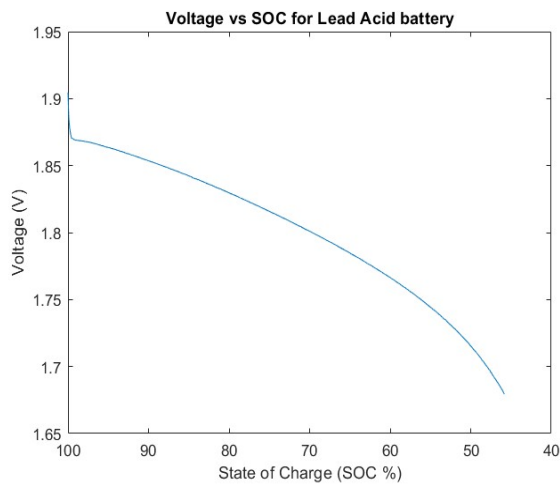
Voltage vs Time:



Lead-Acid Battery: The voltage begins at approximately 1.9 V and gradually decreases over time, with a steady decline indicating a consistent discharge process. The total discharge duration is relatively short, reflecting the limitations of lead-acid batteries in sustaining energy output over time.

Li-Po Battery: The voltage starts at around 4 V and decreases more gradually over an extended period compared to the lead-acid battery. The curve shows a prolonged plateau phase, indicative of stable energy delivery, before a steep drop near the end of the discharge process, typical of Li-Po batteries with higher energy density and efficiency.

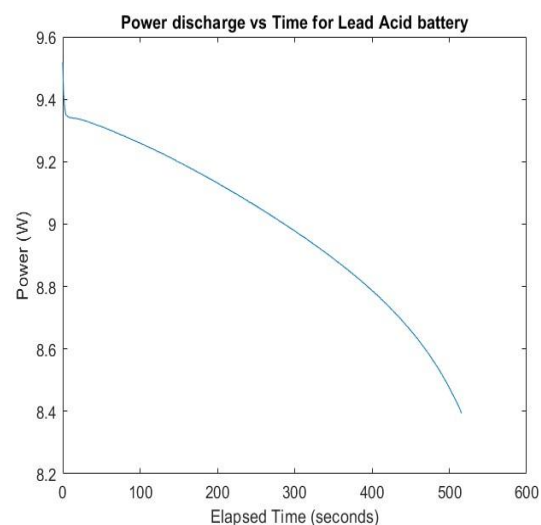
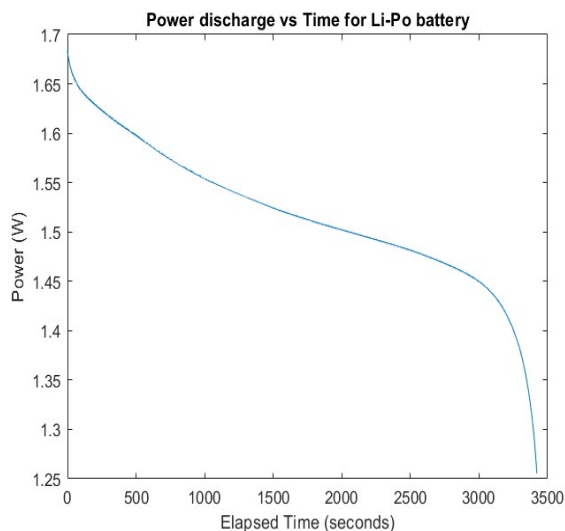
Voltage vs SOC:



Lead-Acid Battery: The voltage begins at approximately 1.9 V when the SOC is 100% and steadily decreases as the SOC reduces. The decline is relatively linear, with the voltage dropping to about 1.7 V when the SOC reaches 50%. This behavior indicates a gradual loss of capacity with a consistent voltage drop during discharge.

Li-Po Battery: The voltage starts at around 4 V at 100% SOC and remains relatively stable in the higher SOC range, showing a plateau phase. However, as the SOC drops below 30%, the voltage decreases rapidly, reaching around 3 V at 0% SOC. This pattern demonstrates the Li-Po battery's ability to maintain a stable voltage for most of its discharge cycle, followed by a sharp drop near depletion.

Power Discharge vs Time

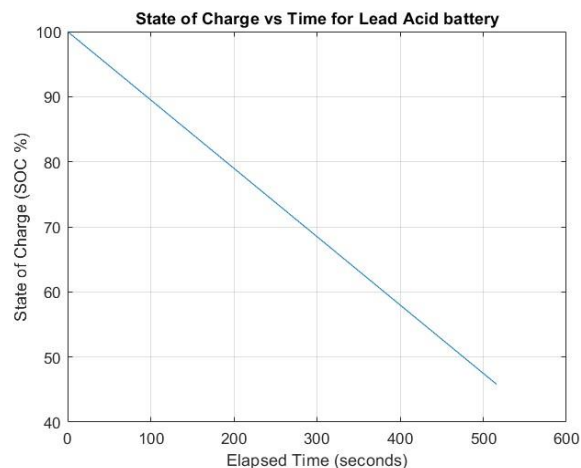
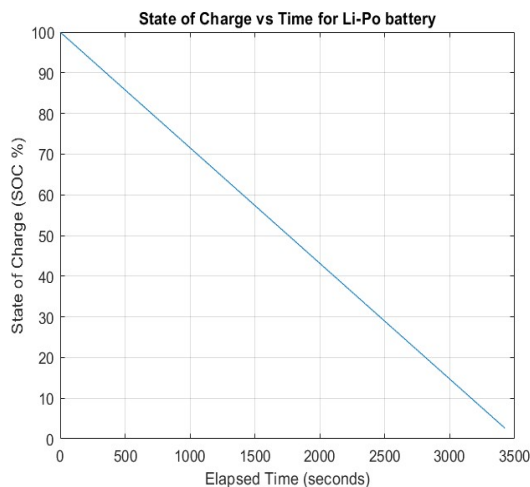


The graphs illustrate the power discharge over time for two different types of batteries: **Li-Po** and **Lead-Acid**.

Li-Po Battery: The power output starts at approximately 1.65 W and decreases gradually over an extended period, maintaining stability throughout most of the discharge cycle. A sharp drop in power occurs near the end, typical of Li-Po batteries as they approach full depletion. This highlights their ability to sustain consistent power delivery over a long duration.

Lead-Acid Battery: The power output begins at around 9.4 W and steadily declines over a shorter duration. The discharge process is more linear, reflecting the battery's lower energy density and faster depletion compared to the Li-Po battery.

State of Charge vs Time



Li-Po Battery: The SOC decreases linearly over an extended period, starting at 100% and reaching 0% after approximately 3,500 seconds. This steady decline indicates a gradual energy depletion and a longer discharge duration, typical of Li-Po batteries with high energy density.

Lead-Acid Battery: The SOC also decreases linearly but over a much shorter timeframe, starting at 100% and reaching 0% after around 500 seconds. This faster depletion reflects the lower energy density and shorter operational duration of Lead-Acid batteries.

4. Conclusion

The battery discharge tests provided valuable insights into the impact of C-rate, temperature, and cell chemistry on battery performance. These experiments revealed key observations about how batteries behave under various operating conditions, which are summarized as follows:

- 1- Impact of C-rate: Lower C-rates (0.5C) allow for extended discharge durations and steady energy output, making them ideal for applications requiring consistent and prolonged power delivery. Conversely, higher C-rates (2C) result in significantly shorter discharge times due to rapid energy depletion and increased internal resistance, better suited for applications demanding high power over shorter intervals.
- 2- Impact of Temperature: High temperatures (+40°C) improve battery performance by reducing internal resistance and stabilizing voltage, enabling longer operational times. On the other hand, low temperatures (-10°C) greatly impair performance by increasing resistance, accelerating energy depletion, and reducing voltage stability.

- 3- Impact of Cell Chemistry: Ni-MH batteries exhibit predictable discharge behavior across various conditions, while lead-acid batteries show a steady power decline. Li-Po batteries stand out for their ability to maintain stable power output until near the end of their discharge cycle. This emphasizes the importance of selecting battery chemistry based on specific application requirements.

In conclusion, understanding the interplay between C-rate, temperature, and cell chemistry is critical for developing efficient and reliable energy storage systems. These findings are particularly relevant as the demand for advanced batteries continues to grow in applications such as electric vehicles, grid storage, and portable electronics.

5. References

- [1] Tuhibur Rahman and Talal Alharbi. Exploring lithium-ion battery degradation: A concise review of critical factors, impacts, data-driven degradation estimation techniques, and sustainable directions for energy storage systems. *Batteries*, 10(7), 2024.
- [2] Alkaya, Alkan; Yüksek, Gökhan. 2023. Effect of the Depth of Discharge and C-Rate on Battery Degradation and Cycle Life. DOI: 10.1109/ELECO60389.2023.10415967
- [3] Monolithic Power Systems. No date. How Battery Characteristics Impact Battery Management. Available at: [How Battery Characteristics Impact Battery Management | Article | MPS](#)